

2-7 Decompose to Subtract

Learning goal: I can break apart the number I am subtracting and take away the parts one at a time to find the difference.

Textbook: pages 59–62. The book subtracts $353 - 184$ by breaking 184 apart in two different ways.

Words to Know

decompose	Break a number into parts. 184 can be $100 + 80 + 4$ or $153 + 30 + 1$.
part	One of the smaller numbers that add up to the whole number.
difference	The answer when you subtract.
decomposition tree	A picture with the number on top and its parts below.
friendly number	A number that is easy to work with, like 200 or 170.

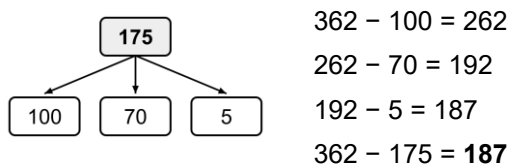
How to Teach It

1. Keep the first number whole. Break apart only the number you are **taking away**.
2. **By place value:** break it into hundreds, tens, and ones. Take away the hundreds, then the tens, then the ones. Each step starts from the last answer.
3. **Another way:** choose a first part that lands you on a friendly number. For $362 - 175$, take away 162 first to land on 200.
4. Check the tree: the parts must **add back** to the number you are taking away.
5. The difference is the same for any correct tree, and the parts can be taken away in any order.

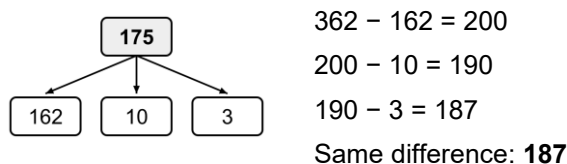
Tip: Ask, “Do your parts add back to the number?” and “Can you land on a hundred?”

Worked Examples

Example 1: Decompose by place value



Example 2: Decompose another way



Example 3: Check the parts

$458 - 102$: Sam breaks 102 into 100 and 20. But $100 + 20 = 120$, not 102.

Use 100 and 2: $458 - 100 = 358$, $358 - 2 = \mathbf{356}$.

Example 4: Any order works

$538 - 215$ with parts 200, 10, 5.

Ones first: 533, 523, $\mathbf{323}$.

Hundreds first: 338, 328, $\mathbf{323}$.

Skills Checklist *Check off each skill your child can do without help.*

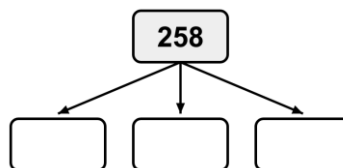
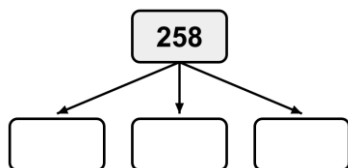
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|--|---|
| ☐ Decompose a 3-digit number in two different ways. | ☐ Check that the parts add back to the number. |
| ☐ Subtract the parts by place value. | ☐ Explain why the difference is the same either way. |
| ☐ Choose parts that land on a friendly number. | ☐ Solve word problems and table problems by subtracting. |

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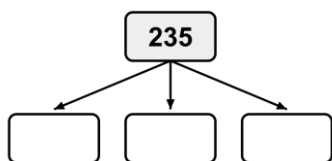
Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Decompose **258** in two different ways.



- B.** Subtract **462 - 235**. Break apart 235 by place value. Take away one part at a time.

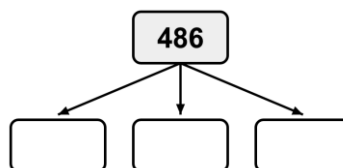
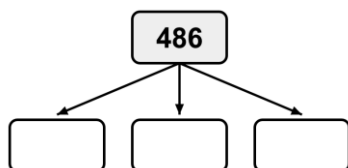


- C.** How can you decompose 263 in two different ways to find **541 - 263**? Show both ways.

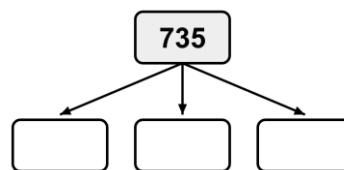
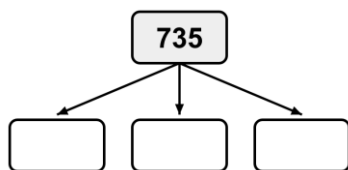
Difference: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Decompose **486** in two different ways.



2. Decompose **735** in two different ways.



3. Decompose one number to subtract.
Why did you choose that way?
 $586 - 241$

Why: _____

4. Decompose one number to subtract.
Why did you choose that way?
 $728 - 460$

Why: _____

5. Find the difference. Show the strategy you used.
 $395 - 172 =$ _____

Difference: _____

6. Find the difference. Show the strategy you used.
_____ $= 442 - 215$

Difference: _____

7. Find the difference. Show the strategy you used.
 $603 - 285 =$ _____

Difference: _____

8. Find the difference. Show the strategy you used.
 $831 - 547 =$ _____

Difference: _____

9. Leo subtracts $576 - 103$. He breaks 103 into 100 and 30. How can you help him understand his mistake? What is the correct difference?

Difference: _____

10. The table shows how many books a class read each day. Find the difference between the **greatest** and **least** number of books.

Day	Books
Monday	214
Tuesday	187
Wednesday	309
Thursday	256

Difference: _____

11. Mia has 357 stickers. 168 are star stickers. The rest are heart stickers. How many heart stickers does she have?

Heart stickers: _____

12. Ana subtracts $649 - 425$. She takes away 5, then 400, then 20. Can she take away the parts in any order? Explain.

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Together

A	Two different trees whose parts add to 258, e.g., 200, 50, 8 and 150, 100, 8 . <i>Accept any 3 parts that add to 258, as long as the two trees use different parts (the same parts in a new order do not count).</i>
B	Tree: 200, 30, 5. $462 - 200 = 262$; $262 - 30 = 232$; $232 - 5 = 227$. Difference: 227
C	Way 1: $263 \rightarrow 200, 60, 3$: $541 - 200 = 341$; $341 - 60 = 281$; $281 - 3 = 278$. Way 2: $263 \rightarrow 241, 22$: $541 - 241 = 300$; $300 - 22 = 278$. Difference: 278 <i>Accept any two correct trees that give 278.</i>

On My Own

1	Sample: 400, 80, 6 and 386, 90, 10 <i>Items 1–2: accept any two trees whose parts add to the number. Reordering the same parts is not a different way.</i>
2	Sample: 700, 30, 5 and 635, 90, 10
3	Sample: $241 \rightarrow 200, 40, 1$: 386, 346, 345 . <i>Reason: place-value parts are easy to take away because no regrouping is needed. Accept other correct trees with a sensible reason.</i>
4	Sample: $460 \rightarrow 428, 32$: $728 - 428 = 300$; $300 - 32 = \mathbf{268}$. Or 400, 60: 328, 268 . <i>Reason: taking away 428 lands on 300, a friendly number. Accept any sensible reason.</i>
5	223 (e.g., $395 - 100 = 295$; $295 - 70 = 225$; $225 - 2 = 223$)
6	227 (e.g., $442 - 200 = 242$; $242 - 10 = 232$; $232 - 5 = 227$)
7	318 (e.g., $603 - 200 = 403$; $403 - 80 = 323$; $323 - 5 = 318$)
8	284 (e.g., $831 - 500 = 331$; $331 - 40 = 291$; $291 - 7 = 284$)
9	$100 + 30 = 130$, not 103. The parts must add back to 103. Use 100 and 3: $576 - 100 = 476$; $476 - 3 = \mathbf{473}$.
10	Greatest 309 (Wed), least 187 (Tue). $309 - 187 = \mathbf{122}$ books. <i>Sample: $309 - 100 = 209$; $209 - 80 = 129$; $129 - 7 = 122$. Watch for subtracting the first and last rows instead.</i>
11	$357 - 168 = \mathbf{189}$ heart stickers <i>Sample: $357 - 100 = 257$; $257 - 60 = 197$; $197 - 8 = 189$.</i>
12	Yes. $649 - 5 = 644$; $644 - 400 = 244$; $244 - 20 = 224$. She still takes away 425 in all, so the difference, 224 , is the same in any order.